Problem Set on Transformers

1. A single-phase, 250-kVA, 11-kV/415-V, 50-Hz transformer has 80 turns on the secondary. Calculate (*a*) the approximate values of the primary and secondary currents, (*b*) the approximate number of primary turns, and (*c*) the maximum value of the flux. [**Ans. :** (*a*) 22.7 A, 602 A; (*b*) 2121; (*c*) 23.4 mWb]

2. The primary winding of a 50-Hz transformer has 480 turns and is fed from a 6400-V supply. Determine (*a*) the peak value of the flux in the core, and (*b*) the secondary voltage if the secondary winding has 20 turns. [**Ans. :** (*a*) 0.06 Wb; (*b*) 266.4 V]

3. A single phase, 50-Hz transformer has 80 turns on the primary winding and 400 turns on the secondary winding. The net cross-sectional area of the core is 200 cm^2. If the primary winding is connected to 240-V, 50-Hz supply, determine (*a*) the emf induced in the secondary winding, and (*b*) the maximum flux density in the core.

[**Ans. :** (*a*) 1200 V; (*b*) 0.675 T]

4. A 10-kVA, single-phase transformer has its primary connected to a 2000-V supply. It has 60 turns on the secondary winding and voltage across it is found to be 240 V. Assuming the transformer to be ideal, calculate (*a*) the number of turns on its primary winding; (*b*) the full-load primary and secondary currents. [**Ans. :** (*a*) 500; (*b*) 5 A, 41.67 A]

5. A 200-kVA, 3300-V/240-V, 50-Hz, single-phase transformer has 80 turns on the secondary winding. Assuming an ideal transformer, calculate (*a*) primary and secondary currents on full load, (*b*) the maximum value of flux, and (*c*) the number of primary turns. [**Ans. :** (*a*) 60.6 A, 833.33 A; (*b*) 0.0135 Wb; (*c*) 1100]

6. A single-phase transformer with 10 : 1 turns-ratio and rated at 50 kVA, 2400-V/240-V, 50-Hz is used to step down the voltage of a distribution system. The low tension (LT) voltage is to be kept constant at 240 V. Find the value of the load impedance of the LT side so that the transformer is loaded fully. Find also the value of the maximum flux inside the core if the LT side has 23 turns. [**Ans. :** 1.152 Ω, 0.047 Wb]

7. The primary of a single-phase transformer takes 1 A at power factor of 0.4 when connected to a 240-V, 50-Hz supply and the secondary is on open circuit. The number of turns on the primary is twice that on the secondary. A load taking 50 A at a lagging power factor of 0.8 is now connected across the secondary. What is now the value of the primary current ? (Neglect the voltage drops in the transformer.)
[**Ans. :** 25.9 A]

8. A single-phase, 50-Hz transformer has 100 turns on the primary winding and 400 turns on the secondary winding. The net cross-sectional area of the core is 250 cm^2. If the

primary winding is connected to a 50-Hz, 230-V supply, calculate (*a*) the emf induced in the secondary winding, and (*b*) the maximum value of the flux density in the core.

[**Ans. :** (*a*) 920 V; (*b*) 0.414 T]

(B) Tricky Problems

9. A single-phase, 50-Hz transformer has 30 turns on primary and 350 turns on secondary. The net cross-sectional area of the core is 250 cm2. If the primary winding is connected to a 230-V, 50-Hz supply, calculate (a) peak value of the flux density in the core, (b) the voltage induced in the secondary winding, and (c) the primary current when the secondary current is 100 A. Neglect the losses.

[Ans. : (a) 1.3814 T; (b) 2683.33 V; (c) 1166.67 A]

10. A single-phase, 50-Hz, 100-kVA, 2400-V/240-V transformer has no-load current of 0.64 A and core loss of 700 W, when its high voltage (HV) side is energized at rated voltage and frequency. Calculate the two components of the no-load current. If this transformer supplies a load current of 40 A at 0.8 lagging pf on its low voltage side, determine the primary current and its power factor. Ignore the resistance and leakage reactance drops.

[Ans. : 0.5697 A, 0.2917 A; 4.584 A, 0.762 lagging]

11. An ideal 50-Hz, core-type transformer has 100 primary-turns and 200 secondary-turns. The primary rated voltage is 220 V. If the maximum permissible flux density is 1.2 T, what should be the cross-sectional area ? Also, find (a) the secondary voltage, and (b) the primary current in complex form with reference to the secondary voltage vector when secondary delivers a current of 8 A at a lagging power factor of 0.8.

[Ans. : ;(a) 440 V; (b)]

12. A 500-kVA, 11000-V/400-V, 50-Hz, single-phase transformer has 100 turns on the secondary winding. Calculate (a) the approximate number of turns on the primary winding, (b) the approximate values of the primary and secondary current, and (c) the maximum value of the flux in the core.

[Ans. : (a) 2750; (b) 45.45 A, 1250 A; (c) 0.018 Wb]